HAMNA ARSHAD

AI/ML | Computer Vision | Python | Generative AI

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Profile

Final-year Computer Science student at UET Lahore with experience in Python, AI/ML, computer vision, and AI-powered applications. Worked on projects involving RAG, object detection, image segmentation, and deep learning.

Skills

Languages
- Python
- C++
- C#
- JavaScript
- TypeScript

Backend & APIs
- Flask
- REST APIs
- API Integration

Web & Software Development
- React.js
- Next.js
- Flask
- HTML
- CSS
- Tailwind CSS
- Bootstrap
- .NET

AI / Application Integration
- OpenAI API
- Gemini API
- RAG
- Embeddings

Tools & Platforms
- Git
- GitHub
- Google Colab
- Odoo

DataBase Management
- PostgreSQL
- SQL
- MongoDB
- Supabase
- pgvector

Programming Concepts
- Object-Oriented Programming
- Data Structures & Algorithms
- Problem Solving
- Database Management
- Data Modeling

AI / Machine Learning
- Machine Learning
- Computer Vision
- Object Detection
- Image Processing
- Feature Extraction
- CNNs
- U-Net / U-Net Lite
- YOLOv8

Education

BS Computer Science, *University Of Engineering And Technology* 2023 – 2027
Continue

FSC Pre-medical (with additional Maths), *Aspire College*

Matriculation
92% Marks, Honours kindergarten

AI / MACHINE LEARNING PROJECTS

Conversa — AI Chatbot Builder, *Next.js, TypeScript, Python, Supabase, PostgreSQL, pgvector, OpenAI*

- Worked on a full-stack AI chatbot builder that allows businesses to create chatbots from their own documents and web content.
- Worked with document and web-content extraction and processing pipelines for preparing knowledge for retrieval.
- Implemented semantic chunking to divide extracted content into smaller knowledge units.
- Worked with a human-in-the-loop workflow for reviewing and validating extracted knowledge before chatbot use.
- Worked with **OpenAI embeddings** and **pgvector** for semantic retrieval.
- Implemented keyword search and hybrid retrieval alongside vector-based search.
- Integrated retrieved information with an LLM-based generation pipeline for conversational responses.

AI-Based Crop Disease Prediction, *Flask, Python, U-Net / U-Net Lite, CNN, OpenCV*

- Developed a three-stage computer vision pipeline for **disease detection, segmentation, and classification**.
- Worked with **20,000+ crop images** covering **9 disease classes**.
- Used OpenCV for image processing and preprocessing.
- Used U-Net/U-Net Lite for disease-region segmentation.
- Used CNN-based classification for crop disease identification.
- Achieved **85.3% test accuracy**.
- Integrated the model pipeline with Flask for real-time diagnosis and disease-severity analysis.

AutoSlides — AI Research Proposal to PowerPoint Generator,
Python, Flask, Gemini API, Document Processing

- Developed an AI-powered application for converting research proposal documents into structured PowerPoint presentations.
- Built a Flask backend for processing requests and coordinating the application workflow.
- Integrated the **Gemini API** for AI-assisted content processing.
- Worked with document processing and automated presentation generation.

Parking Space Detection Using YOLOv8, *Python, YOLOv8, Computer Vision, Google Colab*

- Developed a real-time parking-space detection system using YOLOv8.
- Prepared and worked with **8,500+ annotated images** from the PKLot dataset.
- Trained and evaluated the object-detection model.
- Achieved approximately **99.5% precision / 0.995 mAP@0.5**.
- Achieved approximately **30 ms inference time**.
- Evaluated the model using precision, recall, F1-score, and mAP.

Other PROJECTS

Database Management System, *Database Architecture, Graph Database, Cypher-like Query Processing*

- Developed a graph-based database management system.
- Worked on data modeling and database architecture.
- Implemented a custom Cypher-like query processor.
- Worked on query execution and graph-based data operations.

RESEARCH EXPERIENCE

Parking Space Detection Using YOLOv8, *Undergraduate Research / Experimental Study*

- Prepared and annotated **8,500+ PKLot images** for parking-space detection.
- Trained and evaluated a YOLOv8-based object detection model.
- Evaluated model performance using **precision, recall, F1-score, and mAP**.
- Achieved approximately **0.995 mAP@0.5** with approximately **30 ms inference time**.

- Analyzed model performance for real-time parking detection.

Parallel 2D Heat Diffusion, *Pthreads, OpenMP, MPI*
- Implemented a sequential 2D heat-diffusion solver as a baseline.
- Developed parallel implementations using **Pthreads, OpenMP, and MPI**.
- Conducted comparative performance analysis using strong and weak scaling.
- Analyzed speedup, efficiency, synchronization overhead, and memory-bandwidth bottlenecks.

Certificates

- Introduction to Database and SQL
- MongoDB
- Odoo Web Development

Languages

- English
- Urdu